Confidence Intervals (t-Distribution)

Statistics Worksheet · Grade 11–College

Name: _____ Date: _____

Learning Objectives

- Calculate the point estimate, margin of error, standard error, and sample standard deviation from a given confidence interval
- Determine the critical t-value using degrees of freedom and confidence level
- Write a statistically correct interpretation of a confidence interval in context

Problems

1. A 95% confidence interval for the average lifespan of AA batteries is given as (430, 470) minutes. What is the point estimate (sample mean)?

$$\bar{x} = \frac{\text{Lower Limit} + \text{Upper Limit}}{2}$$

2. Using the confidence interval (430, 470) minutes, calculate the margin of error.

$$E = \text{Upper Limit} - \bar{x}$$

3. A confidence interval is reported as (118, 134). Find the point estimate and the margin of error.

$$\bar{x} = \frac{118 + 134}{2}, \quad E = \frac{134 - 118}{2}$$

4. Find the critical t-value (t*) for a 95% confidence interval with a sample size of 30. Use the formula: alpha divided by 2, where alpha equals 1 minus the confidence level.

$$\alpha/2 = \frac{1 - 0.95}{2} = 0.025, \quad df = n - 1 = 29$$

5. Using the AA battery example with a 95% confidence interval of (430, 470) minutes, a sample size of 30, and a critical t-value of 2.045, find the sample standard deviation.

$$E = t^* \cdot \frac{s}{\sqrt{n}} \Rightarrow s = \frac{E \cdot \sqrt{n}}{t^*}$$

6. Using the AA battery example with a 95% confidence interval of (430, 470) minutes, a sample size of 30, and a critical t-value of 2.045, calculate the standard error.

$$SE = \frac{s}{\sqrt{n}}$$

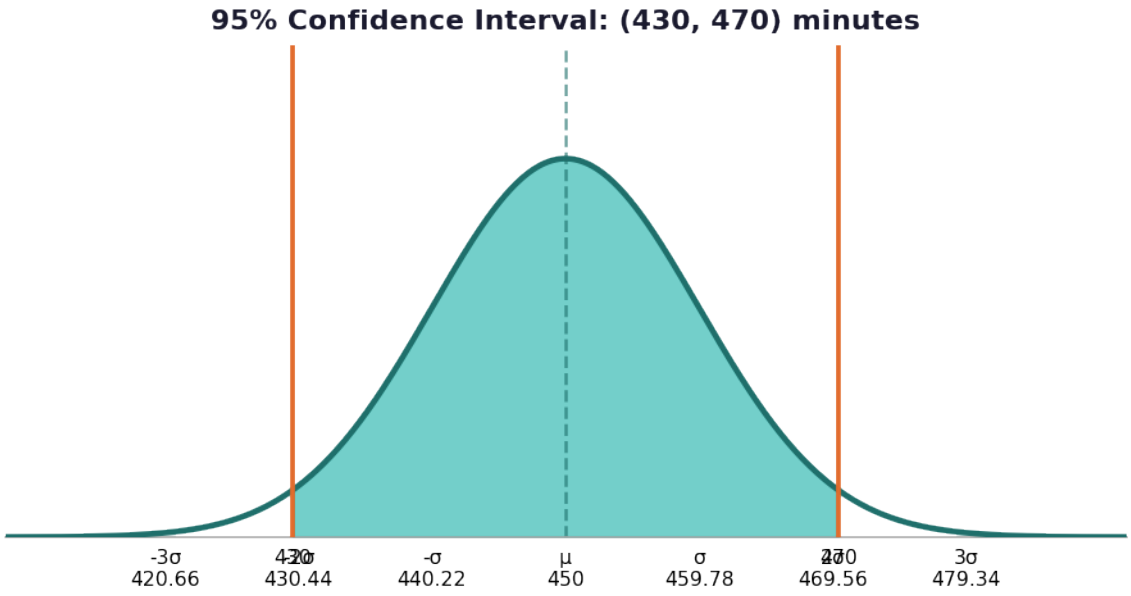
7. A study reports a 90% confidence interval for average sleep duration as (6.4, 7.8) hours based on a sample of 25 people. Find the critical t-value for this interval.

$$\alpha/2 = \frac{1 - 0.90}{2} = 0.05, \quad df = 24$$

8. A 90% confidence interval for average sleep duration is (6.4, 7.8) hours, based on a sample of 25 people. The critical t-value is 1.711. Find the sample standard deviation and the standard error.

$$E = \frac{7.8 - 6.4}{2}, \quad s = \frac{E \cdot \sqrt{n}}{t^*}, \quad SE = \frac{s}{\sqrt{n}}$$

9. Write a statistically correct interpretation of the following confidence interval: A 95% confidence interval for the average lifespan of AA batteries is (430, 470) minutes, based on a sample of 30 batteries.



10. A researcher claims that the average monthly electricity bill in a city is between \$112 and \$148, based on a random sample of 40 households with a 99% confidence level. Find: (a) the point estimate, (b) the margin of error, (c) the critical t-value, (d) the standard error, and (e) the sample standard deviation. Then write a correct interpretation of the interval.

$\bar{x} = \frac{112 + 148}{2}$, $E = \frac{148 - 112}{2}$, $t^* = t_{\alpha/2,\, df = 39}$, $s = \frac{E\sqrt{n}}{t^*}$, $SE = \frac{s}{\sqrt{n}}$

Confidence Intervals (t-Distribution) — Answer Key

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Answer Key

1. Answer: 450 minutes
- Point estimate = midpoint of the confidence interval
- $\bar{x}$ = (430 + 470) / 2 = 900 / 2 = 450 minutes

2. Answer: 20 minutes
- Point estimate $\bar{x}$ = 450
- Margin of error E = 470 – 450 = 20 minutes
- Alternatively, E = (470 – 430) / 2 = 40 / 2 = 20 minutes

3. Answer: Point estimate = 126; Margin of error = 8
- $\bar{x}$ = (118 + 134) / 2 = 252 / 2 = 126
- E = (134 – 118) / 2 = 16 / 2 = 8

4. Answer: t* ≈ 2.045
- α = 1 – 0.95 = 0.05
- $\alpha/2$ = 0.025
- Degrees of freedom = n – 1 = 30 – 1 = 29
- Using inverse-t: t*(0.025, df=29) ≈ 2.045

5. Answer: s ≈ 53.57 minutes
- E = 20, t* = 2.045, n = 30
- s = (E × √n) / t*
- s = (20 × √30) / 2.045
- s = (20 × 5.477) / 2.045 ≈ 109.54 / 2.045 ≈ 53.57 minutes

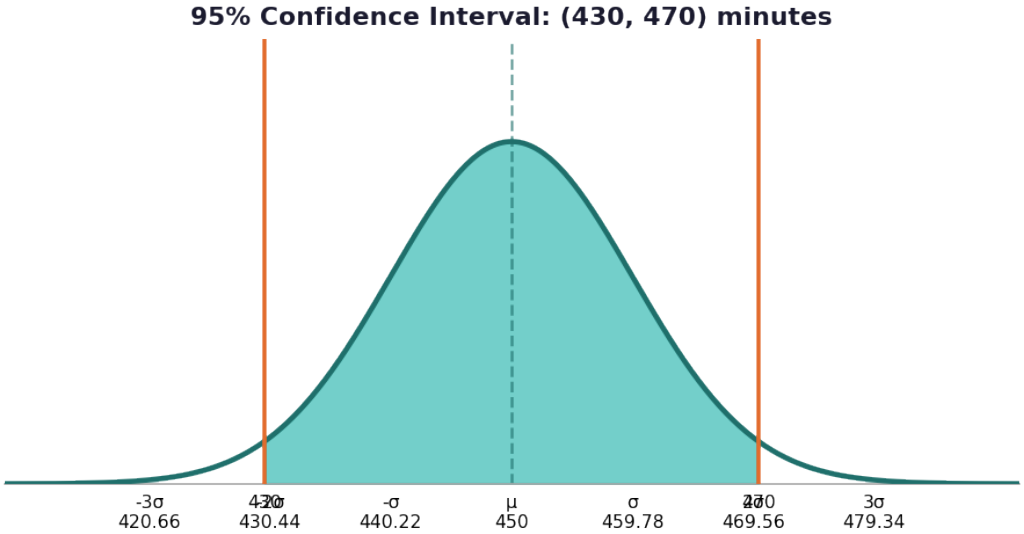
6. Answer: SE ≈ 9.78 minutes
- From Problem 5, s ≈ 53.57
- SE = s / √n = 53.57 / √30
- SE = 53.57 / 5.477 ≈ 9.78 minutes
- Check: t* × SE = 2.045 × 9.78 ≈ 20 ✓

7. Answer: t* ≈ 1.711
- α = 1 – 0.90 = 0.10
- $\alpha/2$ = 0.05
- df = n – 1 = 25 – 1 = 24
- Using inverse-t: t*(0.05, df=24) ≈ 1.711

8. Answer: E = 0.7 hrs; s ≈ 2.047 hrs; SE ≈ 0.409 hrs

- $\bar{x}$ = (6.4 + 7.8)/2 = 7.1; E = 7.8 – 7.1 = 0.7 hours
- s = (E × √n) / t* = (0.7 × √25) / 1.711 = (0.7 × 5) / 1.711 = 3.5 / 1.711 ≈ 2.047 hours
- SE = s / √n = 2.047 / 5 ≈ 0.409 hours
- Check: t* × SE = 1.711 × 0.409 ≈ 0.7 ✓

9. Answer: We are 95% confident that the true mean lifespan of all AA batteries falls between 430 and 470 minutes.



- Identify the confidence level: 95%
- Identify the parameter: the TRUE POPULATION mean lifespan of AA batteries
- Identify the interval bounds: 430 to 470 minutes
- Correct interpretation: 'We are 95% confident that the true mean lifespan of AA batteries is between 430 and 470 minutes.' — This does NOT mean there is a 95% probability that a single battery lasts in this range.

10. Answer: $\bar{x}$ = $130; E = $18; t* ≈ 2.708; SE ≈ 6.65; s ≈ 42.04

- (a) $\bar{x}$ = (112 + 148) / 2 = 260 / 2 = 130
- (b) E = 148 – 130 = 18
- (c) α = 1 – 0.99 = 0.01; $\alpha/2$ = 0.005; df = 40 – 1 = 39; t*(0.005, df=39) ≈ 2.708
- (d) SE = E / t* = 18 / 2.708 ≈ 6.65
- (e) s = SE × √n = 6.65 × √40 = 6.65 × 6.325 ≈ 42.04
- Interpretation: We are 99% confident that the true mean monthly electricity bill for all households in the city is between $112 and $148.